جامعة الشارقة
UNIVERSITY OF SHARJAH

College of Computing and Informatics
Department of Computer Engineering

Course Syllabus

Course Title
Digital Logic Design Laboratory

Course Code
1502202

Updated on Summer Semester, 2022/2023

Course Syllabus

Course Title	Digital Logic Design Laboratory
Course Code	1502202
Credit Hours	1
Pre-requisite(s)	1502201 Digital Logic Design
Pre/Co-requisite(s)	None
Semester	Summer | **Year** | 2022/2023
Instructors Name	Prof. Ibrahim Kamel
Office Location	
Tel. No.	06-5050931
Email	Kamel@sharjah.ac.ae
Lecture Times	Monday/Wednesday 14:00 - 16:45
Office Hours	

Course Description (as in the catalogue):

This course presents the operation of basic logic gates as well as some combinational and sequential circuits. The course illustrates how to design and implement different logic circuits such as adders, subtractors, decoders, encoders, flip-flops, counters, and shift registers. The design and implementation of the logic circuits will be accomplished practically in the lab by using Hardware Description Language (HDL) and circuit boards.

Course Learning Outcomes (Course SLOs):

By the end of successful completion of this course, the student will be able to:

1. Apply digital logic design procedures to implement digital logic circuits that solves engineering problems.
2. Work in groups to design and build practical digital circuits.
3. Practice troubleshooting of the designed circuits to verify their behavior.
4. Describe digital circuits using Hardware Description Language.

Alignment of Course Student Learning Outcomes to BSCPE Program Student Learning Outcomes

BSCPE Program/Student Learning Outcomes (PLOs/SLOs)	Course SLOs
1. Identify, formulate, and solve complex computer engineering problems by applying principles of engineering, science, and mathematics.	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 3-Practice troubleshooting of the designed circuits to verify their behavior. 4-Describe digital circuits using Hardware Description Language.
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 4-Describe digital circuits using Hardware Description Language.
3. Communicate effectively with a range of audiences	2-Work in groups to design and build practical digital circuits.
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	2-Work in groups to design and build practical digital circuits.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 3-Practice troubleshooting of the designed circuits to verify their behavior. 4-Describe digital circuits using Hardware Description Language.

Alignment of Course Student Learning Outcomes to BSCSE Program Student Learning Outcomes

BSCSE Program/Student Learning Outcomes (PLOs/SLOs)	Course SLOs
1. Identify, formulate, and solve complex problems in cybersecurity engineering or related fields by applying principles of engineering, science, and mathematics.	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 3-Practice troubleshooting of the designed circuits to verify their behavior. 4-Describe digital circuits using Hardware Description Language.
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 4-Describe digital circuits using Hardware Description Language.
3. Communicate effectively with a range of audiences	2-Work in groups to design and build practical digital circuits.
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	2-Work in groups to design and build practical digital circuits.
6. Conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1- Apply digital logic design procedures to implement digital logic circuits that solves engineering problems. 3-Practice troubleshooting of the designed circuits to verify their behavior. 4-Describe digital circuits using Hardware Description Language.

Scheduling of laboratory and other non-lecture sessions, including online sessions, as appropriate

Week	Topic	Date	Course SLOs	Tools
1.	Experiment#1: Introduction to Digital Logic Design Lab and Design Combinational Circuits Using Basic TTL Gates.	29-May-2023	1,2,3	Circuit Maker Software (CAD Tool)
	Experiment#2: Arithmetic Logic Unit and Data path Utilizing Decoders and Encoders.	31-May-2023	1, 2, 3	Circuit Maker Software (CAD Tool)
2.	Experiment#3: Design Sequential Circuits. **+ Project topic submission**	05-June-2023	1, 2, 3, 4	Circuit Maker Software (CAD Tool)
	Experiment#4: Registers and Counters with Design Applications. **(Quiz-1)**	07-June-2023	1, 2, 3	Circuit Maker Software (CAD Tool)
3.	**MIDTERM EXAM (ON CAMPUS)**	12-June-2023	1, 2, 3	
	Experiment#5: Introduction to Hardware Description Language and Synthesis with FPGAs	14-June-2023		Verilog – HDL and Quartus II 13.1 Software
4.	Experiment#6: Implementation of Combinational Designs Using Verilog Hardware Description Language and Synthesis with FPGAs.	19-June-2023	1, 2, 3, 4	Verilog – HDL and Quartus II 13.1 Software
	Experiment#7: Implementation of Sequential Designs Using Verilog Hardware Description Language and Synthesis with FPGAs. **(Quiz-2)**	21-June-2023	1, 2, 3, 4	Verilog – HDL and Quartus II 13.1 Software
5.	Experiment#8: Practice Implementation of Digital Designs Using Hardware Description Language and Synthesis with FPGAs **+ Project Progress Discussion#1**	26-June-2023	1,2,3,4	Verilog – HDL and Quartus II 13.1 Software
	EID Al ADHA HOLIDAYS			
6.	Practice problems + **Project Discussion #2**	03-July-2023	1, 2, 3, 4	Verilog – HDL and Quartus II 13.1 Software
	Project Demo	05-July-2023	1, 2, 3, 4	
7.	**FINAL EXAM (ON CAMPUS)**	**10-July-2023**		**Include All Experiments**

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Reports	Weekly Submission	5
Quizzes	Quiz1 – (07-June-2023) Quiz2 – (21-June-2023)	20
Performance		5
Midterm	Week 3	25
Project	Topic Submission - (05-June-2023) Project Discussion#1 - (26-June-2023) Project Discussion#2 - (03-July-2023) Project Demo - (05-July-2023)	15
Final	Week 7	30
Total		**100**

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course Los	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
1. Apply digital logic design procedures to implement digital logic circuits that meet certain specifications.	- Presentations - Laboratory Experiment - Guided design projects	- Reports - Quizzes - Exams - Project	70%
2. Work in groups to conduct laboratory experiments.	- Demonstrations - Laboratory Experiment	- Reports - Project	70%
3. Practice troubleshooting of the designed circuits to verify their behavior.	- Laboratory Experiment - Team discussions	- Reports - Quizzes - Exams - Project	70%
4. Design digital circuits using Hardware Description Language.	- Presentations - Team discussions - Laboratory Experiment	- Reports - Exams	70%

Practical Component – Laboratory

There is an accompanying, but separate, theory course (1502201 – Digital Logic Design). The theory course is a prerequisite for the laboratory course. The student is expected to complete the theory course before enrolling in the laboratory course. The laboratory component will give practical application of the theoretical concepts learned in the theory course.

Teaching and Learning Resources:

Text Book(s):

Recommended Readings:

Digital Design With an Introduction to the Verilog HDL, VHDL, and System Verilog
M. Morris Mano, Michael D. Ciletti
Sixth edition, 2017
ISBN-13: 978-0134549897

Other Resources:

Computer Usage

Students use software such as Verilog development tools, CircuitMaker to simulate some of the combinational and sequential circuits they design.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Makeup Exam Policy

Makeup exams are not given for quizzes and midterms.

If a student misses the midterm exam with a legitimate excuse and verifiable evidence, they weight of the midterm exam will be shifted to the final exam.

Grading Rubrics for Assessed Elements:

Grading Rubric for Course Project (Technical)								
Scaled Score:		**F(0)**	**D(1)**	**C(2)**	**C+(3)**	**B(4)**	**A(5)**	
	Weights	**Unacceptable**	**V.Poor**	**Poor**	**Average**	**Good**	**Excellent**	**Weighted Score**
Contribution	25%	No idea of what requirements and objectives are	No more than half of the requirements are missing	Many requirements and objectives are not identified, evaluated and/or completed	All requirements are identified and evaluated but some objectives are not completed	No more than 1 missing elements	All requirements and objectives are identified, evaluated and competed.	
Team Work	25%	No team work was demonstrated	Lot of issues working in a team	Team did not collaborate or communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted.	The team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Members were mostly respectful of each other	The team worked well together to achieve objectives. Not more than one issue.	The team worked well together to achieve objectives. Each member contributed in a valuable way to the project. All data sources indicated a high level of mutual respect and collaboration	

Grading Rubric for Course Project (Technical)								
Scaled Score:		**F(0)**	**D(1)**	**C(2)**	**C+(3)**	**B(4)**	**A(5)**	
	Weights	**Unacceptable**	**V.Poor**	**Poor**	**Average**	**Good**	**Excellent**	**Weighted Score**
Subject knowledge	20%	No understanding of the subject	Serious issues with the understanding	The deliverable did not demonstrate knowledge of the course content, evidence of the research effort or depth of thinking about the topic	The deliverable demonstrated knowledge of the course content by integrating major concepts into the response. The deliverable also demonstrated evidence of limited research effort and/or initial of thinking about the topic	Not more than one serious issue	The deliverable demonstrated knowledge of the course content by integrating major and minor concepts into the response. The deliverable also demonstrated evidence of extensive research effort and a depth of thinking about the topic.	
Supporting Material	20%	No supporting material	Very little supporting material	Insufficient information was obtained and/or sources lack validity. Analysis and design considerations were not supported by the information collected	Sufficient information was obtained and most sources were valid. Analysis and design considerations were mostly supported by the information.	Not more than one issue with regards to supporting material	All relevant information was obtained and information sources were valid. Analysis and design considerations were well supported by the information	

Grading Rubric for Course Project (Technical)								
Scaled Score:		**F(0)**	**D(1)**	**C(2)**	**C+(3)**	**B(4)**	**A(5)**	
	Weights	**Unacceptable**	**V.Poor**	**Poor**	**Average**	**Good**	**Excellent**	**Weighted Score**
Composition	10%	No structure	Almost impossible to understand the structure and composition of the report	The deliverable lacked overall organization. The reader had to make considerable effort to understand the underlying logic and flow of ideas. Diagrams were absent or inconsistent with the text. Grammatical and spelling errors made it difficult for the reader to interpret the text in places	The deliverable was organized and clearly written for the most part. In some areas the logic and/or flow of ideas were difficult to follow. Words were well chosen with some minor expectations Diagrams were consistent with the text. Sentences were mostly grammatical and/or only a few spelling errors were present but they did not hinder the reader.	Not more than 3 minor issues	The deliverable was well organized and clearly written. The underlying logic was clearly articulated and easy to follow. Words were chosen that precisely expressed the intended meaning and supported reader comprehension. Diagrams or analyses enhanced and clarified presentation of ideas. Sentences were grammatical and free from errors.	
Total Score	100%							
Scaled Score (divided by *)								

Comments:

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